

Jupyter Notebook Execution Report

Name: Group 6
Project Title: Lab1
Date: September 18, 2026

Cell 1: ■ Code

```
import numpy as np
import cv2 as cv
from matplotlib import pyplot as plt

img = cv.imread('/home/martin/turtlebot3_ws/src/lab1/ben.jpg')

assert img is not None, "file could not be read, check with os.path.exists()"

gray_img = cv.cvtColor(img, cv.COLOR_BGR2GRAY)
img_rgb = cv.cvtColor(img, cv.COLOR_BGR2RGB)
plt.subplot(121),plt.imshow(img_rgb)
plt.title('Original Image'), plt.xticks([]), plt.yticks([])
plt.subplot(122),plt.imshow(gray_img, cmap='grey')
plt.title('Greyscale Image'), plt.xticks([]), plt.yticks([])
plt.show()
```

Output:

[STDERR]

<string>:1: UserWarning: FigureCanvasAgg is non-interactive, and thus cannot be shown

Original Image



Greyscale Image



Cell 2: ■ Code

```
img = cv.imread('ben.jpg', cv.IMREAD_GRAYSCALE)
assert img is not None, "file could not be read call for help"
edges = cv.Canny(img, 100, 200)

plt.subplot(121),plt.imshow(img,cmap = 'gray')
plt.title('Original Image'), plt.xticks([]), plt.yticks([])
plt.subplot(122),plt.imshow(edges,cmap = 'gray')
plt.title('Edge Image'), plt.xticks([]), plt.yticks([])
plt.show()
```

Original Image



Edge Image



Cell 3: ■ Code

```
import cv2 as cv
face_cascade = cv.CascadeClassifier(cv.data.harcascades +
'haarcascade_frontalface_default.xml')

faces = face_cascade.detectMultiScale(gray_img, scaleFactor=1.1, minNeighbors=5)
for (x,y,w,h) in faces:
cv.rectangle(img, (x,y), (x+w, y+h), (255,0,0), 2)

plt.imshow(cv.cvtColor(img, cv.COLOR_BGR2RGB))
```

Output:

```
<matplotlib.image.AxesImage object at 0x7b47669e4370>
```

